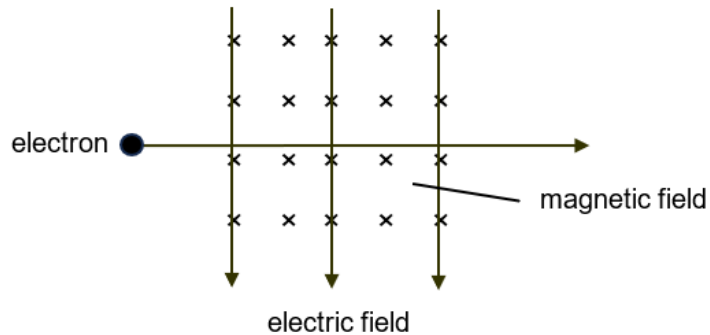


24 A beam of electrons enters a region in which there are magnetic and electric fields directed at right angles to each other and to the beam direction as shown below. The electron beam passes straight through without deflection.



A second beam of electrons travelling at half the speed of the first beam of electrons is then directed along the same line.

How is this second beam deviated?

- | | |
|---|---------------------------------|
| A | Upwards in the plane of paper |
| B | Downwards in the plane of paper |
| C | Out of the plane of the paper |
| D | Into the plane of the paper |

L2 Answer: A

The electric force acting on the electrons is directed upwards in the plane of the paper.
Electric force = qE , which is independent of speed.

By Fleming's left hand rule, the magnetic force acting on the electrons is directed downwards in the plane of the paper.

Magnetic force = Bqv , which is proportional to speed.

When the speed of the second beam of electrons is half its original speed, the magnetic force decreases (halved) while the electric force remains unchanged. Thus, the second beam of electrons deviates upwards in the plane of the paper.

25 A 20 turns square coil of side of length $x = 8.0$ mm is pivoted at its centre and placed in a